**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 076 · Defensive Design and Maintainability

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does authentication establish?

Q2. What is POP used for?
A. Sending new email
B. Transferring web pages
C. Looking up IP addresses
D. Downloading email

Q3. “Changing one protocol layer always requires every other layer to be redesigned.” Correct this. Explain why.

Q4. A drive with private records is lost. “Encryption stops files being stolen or deleted.” Correct this. What does it protect?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Meaningful naming — Descriptive identifiers such as totalCost make code easier to understand.

Indentation — Consistent indentation reveals which instructions belong in a block.

Comments — Explain important intent, rather than repeating obvious syntax line by line.

Subprograms — Splitting code into related functions or procedures helps reuse, testing and maintenance.

Test case — An input and expected outcome used to check correctness.

Bug — A defect in the program.

## New learning

Meaningful naming: Descriptive identifiers such as totalCost make code easier to understand. Consider this in a library software.

Indentation: Consistent indentation reveals which instructions belong in a block. Consider this in a ticketing program.

Comments: Explain important intent, rather than repeating obvious syntax line by line. Consider this in a school marking tool.

Subprograms: Splitting code into related functions or procedures helps reuse, testing and maintenance. Consider this in a shop checkout.

## Worked example

A library software needs meaningful naming and indentation. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1].

3. Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1].

4. Give a specific point and a linked reason for each feature.

# We Do · Practise with guidance

### Guided 1

In a ticketing program, explain the purpose of meaningful naming. [2 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1].

### Guided 2

A school marking tool needs indentation. Explain its role. [2 marks]

Worked method: Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1].

### Guided 3

A shop checkout uses comments and subprograms. Explain both. [4 marks]

Worked method: Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. Subprograms [1]; Splitting code into related functions or procedures helps reuse, testing and maintenance. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Add a comment explaining what the subtraction calculates.

```
originalKB = 500
compressedKB = 300
// add your comment here
savedKB = originalKB - compressedKB
print(savedKB)
```

write

### Question 1

State TWO concepts relevant to this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

### Question 2

Explain how indentation and comments help in a travel app. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

### Question 3

Explain the roles of meaningful naming, comments and subprograms in a club scoring. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

Descriptive identifiers such as totalCost make code easier to understand.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Whether a user has supplied suitable evidence of the claimed identity.

Q2. D — Downloading email

Q3. Layers split communication into tasks. They provide services to each other and help compatibility, development and finding faults.

Q4. Without the key, an attacker cannot easily read ciphertext. Encryption does not stop theft or deletion. Use other controls for other threats.

## I Do / We Do pages · Guided practice

Q1. Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1].

Q2. Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1].

Q3. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. Subprograms [1]; Splitting code into related functions or procedures helps reuse, testing and maintenance. [1].

## You Do pages · Code First

```
originalKB = 500
compressedKB = 300
// Work out how much storage compression saved.
savedKB = originalKB - compressedKB
print(savedKB)
```

Expected result: 200

## You Do pages · Independent questions

Q1. Meaningful naming [1]; Comments [1].

Q2. Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1]. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q3. Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1]. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. Subprograms [1]; Splitting code into related functions or procedures helps reuse, testing and maintenance. [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**